

Chapter 7

Derivatives of Trigonometric Functions

§ Section: 3.3

7.1 Special Limits of Sine and Cosine

Example 1.1

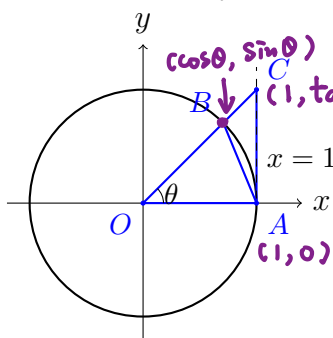
Try to compute $\frac{d}{dx} \sin x$ from the definition and we encounter some special limits.

Hint:
$$\begin{cases} \sin(a+b) = \sin a \cos b + \cos a \sin b \\ \cos(a+b) = \cos a \cos b - \sin a \sin b \end{cases}$$

$$\begin{aligned} \text{Sol: } \frac{d}{dx} \sin x &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = \lim_{h \rightarrow 0} \frac{1}{h} [\sin x \cos h + \cos x \sin h - \sin x] \\ &= \lim_{h \rightarrow 0} \sin x \frac{(\cos h - 1)}{h} + \cos x \cdot \frac{\sin h}{h} = \cos x \\ &= \sin x \left(\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} \right) + \cos x \left(\lim_{h \rightarrow 0} \frac{\sin h}{h} \right) \end{aligned}$$

Example 1.2

Compute $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$.



$$\therefore \triangle OAB \subset \triangle OAC$$

$$\therefore A(\triangle OAB) < A(\triangle OAB) < A(\triangle OAC)$$

$$\Rightarrow \frac{1}{2} \sin \theta < \frac{\theta}{2} \cdot 1^2 < \frac{1}{2} \tan \theta = \frac{1}{2} \frac{\sin \theta}{\cos \theta}$$

$$\text{For } \theta > 0 \Rightarrow \frac{\sin \theta}{\theta} < 1 \quad \frac{\sin \theta}{\theta} > \cos \theta$$

$$\Rightarrow \cos \theta < \frac{\sin \theta}{\theta} < 1$$

By the Squeeze Thm $\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1$.

Moreover, $f(\theta) = \frac{\sin \theta}{\theta}$ is an even function (i.e. $f(-\theta) = f(\theta)$).

Hence $\lim_{\theta \rightarrow 0^-} \frac{\sin \theta}{\theta} = \lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1$. Therefore, $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$.

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x$$

Example 1.3

Compute $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x}$ Sol 1:

$$= \lim_{x \rightarrow 0} \frac{(1 - 2 \sin^2 \frac{x}{2}) - 1}{x} = - \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\frac{x}{2}} = - \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \cdot \sin \frac{x}{2} = -1 \cdot 0 = 0$$

Sol 2:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} &= \lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x + 1)}{x(\cos x + 1)} = \lim_{x \rightarrow 0} \frac{-\sin^2 x}{x(\cos x + 1)} \\ &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{-\sin x}{\cos x + 1} = 1 \cdot 0 = 0 \end{aligned}$$

Example 1.4

Compute $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2}$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x + 1)}{x^2(\cos x + 1)} \\ &= \lim_{x \rightarrow 0} \frac{-\sin^2 x}{x^2(\cos x + 1)} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \left(\frac{-1}{\cos x + 1} \right) \\ &= 1^2 \cdot \left(-\frac{1}{2} \right) = -\frac{1}{2} \end{aligned}$$

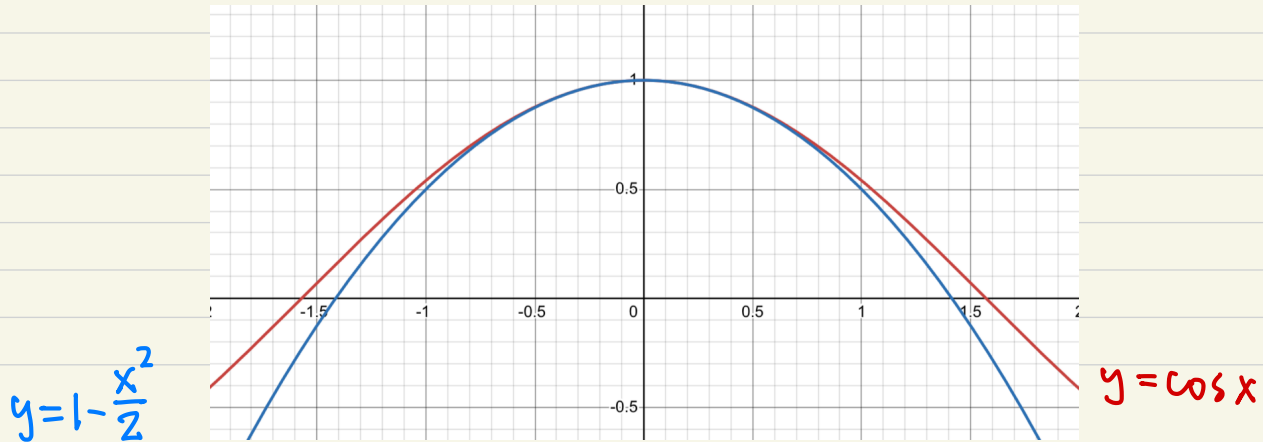
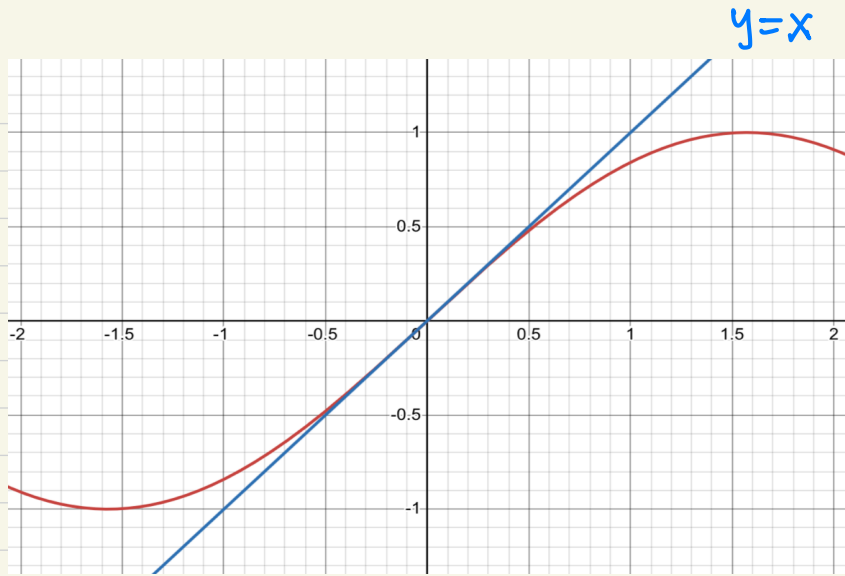
$$\Rightarrow \cos x - 1 \approx -\frac{1}{2}x^2 \Rightarrow \cos x \approx 1 - \frac{1}{2}x^2 \text{ when } x \text{ is near } 0.$$

Special Limits:

1. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ ✓ $\Rightarrow \sin x \approx x$ when x is near 0.

2. $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$

3. $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} = -\frac{1}{2}$ $\Rightarrow \cos x \approx 1 - \frac{1}{2}x^2$ when x is near 0



Now we use $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ to find other limits.

Example 1.5

Find $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$, where $a, b \neq 0$.

Sol: $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \lim_{x \rightarrow 0} \frac{bx}{\sin bx} \cdot \frac{a}{b} = 1 \cdot 1 \cdot \frac{a}{b} = \frac{a}{b}$

Example 1.6

Find $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{(\tan x)^2} = \lim_{x \rightarrow 0} \frac{(1 - \cos 3x)(1 + \cos 3x)}{(\tan x)^2 (1 + \cos 3x)}$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 3x}{\sin^2 x \left(\frac{1 + \cos 3x}{(\cos x)^2} \right)} = \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{\sin x} \right)^2 \cdot \left(\frac{\cos^2 x}{1 + \cos 3x} \right)$$

$$= 9 \cdot \frac{1}{2} = \frac{9}{2}$$

$$1 - \cos 2x = 2 \sin^2 x$$

Example 1.7**DNE**

Find $\lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos 2x}} = \lim_{x \rightarrow 0} \frac{x \sqrt{1 + \cos 2x}}{\sqrt{1 - \cos 2x} \sqrt{1 + \cos 2x}}$

$$= \lim_{x \rightarrow 0} \frac{x \sqrt{1 + \cos 2x}}{\sqrt{\sin^2 2x}} = \lim_{x \rightarrow 0} \frac{x \sqrt{1 + \cos 2x}}{|\sin 2x|}$$

$$= \lim_{x \rightarrow 0} \frac{x}{|\sin 2x|} \sqrt{1 + \cos 2x}$$

$$\lim_{x \rightarrow 0^+} \frac{x}{|\sin 2x|} \sqrt{1 + \cos 2x}$$

$$= \lim_{x \rightarrow 0^+} \frac{x}{\sin 2x} \sqrt{1 + \cos 2x} = \frac{1}{2} \cdot \sqrt{2} = \frac{\sqrt{2}}{2}$$

$$\lim_{x \rightarrow 0^-} \frac{x}{|\sin 2x|} \sqrt{1 + \cos 2x}$$

$$= \lim_{x \rightarrow 0^-} \frac{x}{-\sin 2x} \sqrt{1 + \cos 2x} = -\frac{\sqrt{2}}{2}$$

Example 1.8

Find $\lim_{x \rightarrow -\infty} \sqrt{4x^2 + 1} \sin \left(\frac{1}{x} \right)$.

$$\begin{aligned}
 \text{sol: } \lim_{x \rightarrow -\infty} \sqrt{4x^2+1} \sin\left(\frac{1}{x}\right) &= \lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2+1}}{x} \cdot \frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}} \\
 &= \lim_{x \rightarrow -\infty} \frac{-2x \cdot \sqrt{1+\frac{1}{4x^2}}}{x} \cdot \frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}} = -2 \cdot 1 = -2.
 \end{aligned}$$

7.2 Derivatives of Trigonometric Functions

Example 2.1

Compute $\frac{d}{dx} \sin x$. $= \cos x$

Example 2.2

Compute $\frac{d}{dx} \cos x$.

$$\begin{aligned}
 \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} &= \lim_{h \rightarrow 0} \frac{(\cos x \cos h - \sin x \sin h) - \cos x}{h} \\
 &= \lim_{h \rightarrow 0} \cos x \left(\frac{\cos h - 1}{h} \right) - \sin x \left(\frac{\sin h}{h} \right) = -\sin x.
 \end{aligned}$$

Example 2.3

Compute $\frac{d^n}{dx^n} \sin x$, $\frac{d^n}{dx^n} \cos x$, where $n \in \mathbb{N}$.

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\sin x)'' = -\sin x$$

$$(\cos x)'' = -\cos x$$

$$(\sin x)''' = -\cos x$$

$$(\cos x)''' = \sin x$$

$$(\sin x)^{(4)} = \sin x$$

$$(\cos x)^{(4)} = \cos x$$

Example 2.4

Compute $\frac{d}{dx} \tan x = \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) = \frac{(\sin x)' \cdot \cos x - \sin x \cdot (\cos x)'}{\cos^2 x}$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

Example 2.5

Compute $\frac{d}{dx} \sec x = \frac{d}{dx} \left(\frac{1}{\cos x} \right) = \frac{-(\cos x)'}{\cos^2 x} = \frac{\sin x}{\cos^2 x}$

$$= \tan x \cdot \sec x.$$

Derivatives of Trigonometric Functions:

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

Example 2.6

Compute $\frac{d^2}{dx^2}(\sec x)$.

Sol: $\frac{d}{dx} \sec x = \sec x \cdot \tan x$

$$\begin{aligned} \frac{d^2}{dx^2} \sec x &= (\sec x \cdot \tan x) \tan x + \sec x \cdot (\sec x)^2 \\ &= \sec x \cdot \tan^2 x + \sec^3 x. \end{aligned}$$